

Important: Text generated by AI and may contain incorrect information, always check the content.
To analyze the main characteristics of the dataset, we first need to check its structure and content.

```
```python
Import necessary libraries
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

Check the shape of the dataframe
display('DataFrame Shape:')
display(df.shape)

Check the data types of the columns
display('Data Types:')
display(df.dtypes)

Check for missing values
display('Missing Values:')
display(df.isnull().sum())

Display summary statistics for numerical columns
display('Summary statistics for numerical columns:')
display(df.describe())

Check the distribution of the target variable 'fetal_health'
display('Distribution of fetal_health:')
display(df['fetal_health'].value_counts())

Plot a histogram for the target variable 'fetal_health'
plt.figure(figsize=(8,6))
sns.countplot(x='fetal_health', data=df)
plt.title('Distribution of fetal_health')
plt.show()
```

'DataFrame Shape:'
(2126, 22)
'Data Types:'
```

| | 0 |
|--|---------|
| baseline value | float64 |
| accelerations | float64 |
| fetal_movement | float64 |
| uterine_contractions | float64 |
| light_decelerations | float64 |
| severe_decelerations | float64 |
| prolongued_decelerations | float64 |
| abnormal_short_term_variability | float64 |
| mean_value_of_short_term_variability | float64 |
| percentage_of_time_with_abnormal_long_term_variability | float64 |
| mean_value_of_long_term_variability | float64 |





| 0 | |
|----------------------------|---------|
| histogram_width | float64 |
| histogram_min | float64 |
| histogram_max | float64 |
| histogram_number_of_peaks | float64 |
| histogram_number_of_zeroes | float64 |
| histogram_mode | float64 |
| histogram_mean | float64 |
| histogram_median | float64 |
| histogram_variance | float64 |
| histogram_tendency | float64 |
| fetal_health | float64 |

dtype: object
'Missing Values:'

| 0 | |
|--|---|
| baseline value | 0 |
| accelerations | 0 |
| fetal_movement | 0 |
| uterine_contractions | 0 |
| light_decelerations | 0 |
| severe_decelerations | 0 |
| prolongued_decelerations | 0 |
| abnormal_short_term_variability | 0 |
| mean_value_of_short_term_variability | 0 |
| percentage_of_time_with_abnormal_long_term_variability | 0 |
| mean_value_of_long_term_variability | 0 |
| histogram_width | 0 |
| histogram_min | 0 |
| histogram_max | 0 |
| histogram_number_of_peaks | 0 |
| histogram_number_of_zeroes | 0 |
| histogram_mode | 0 |





| | 0 |
|--------------------|---|
| histogram_mean | 0 |
| histogram_median | 0 |
| histogram_variance | 0 |
| histogram_tendency | 0 |
| fetal_health | 0 |

dtype: int64
'Summary statistics for numerical columns:'

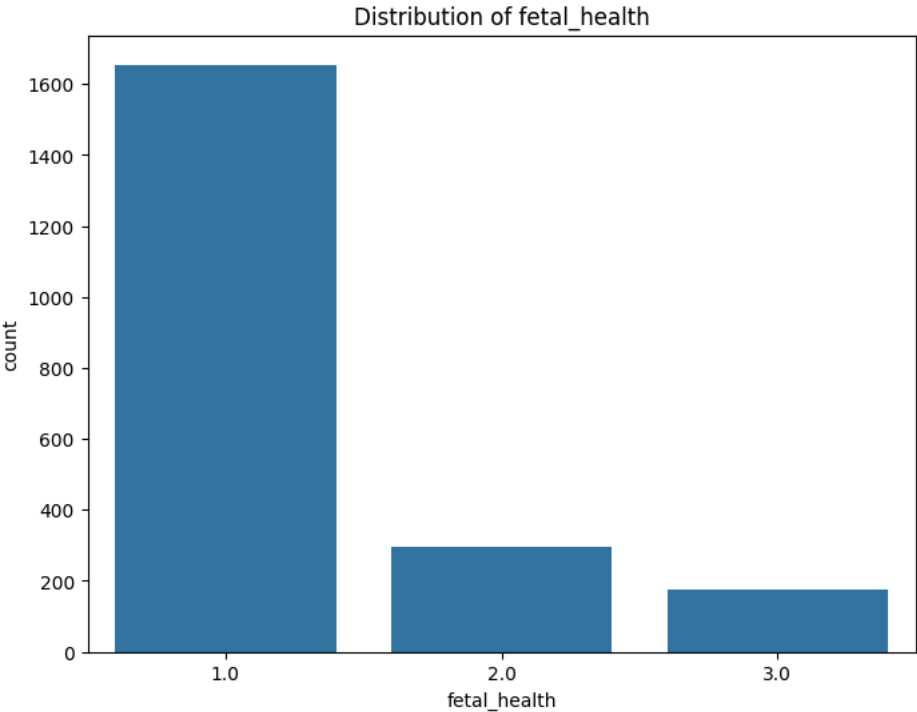
| | baseline
value | accelerations | fetal_movement | uterine_contractions | light_decelerations | severe_decelerations | prolongued_decelerations | abnormal_short_term_variability | mean_value_of_short_term_variability | pe |
|-------|-------------------|---------------|----------------|----------------------|---------------------|----------------------|--------------------------|---------------------------------|--------------------------------------|-------------|
| count | 2126.000000 | 2126.000000 | 2126.000000 | 2126.000000 | 2126.000000 | 2126.000000 | 2126.000000 | 2126.000000 | 2126.000000 | 2126.000000 |
| mean | 133.303857 | 0.003178 | 0.009481 | 0.004366 | 0.001889 | 0.000003 | 0.000159 | 46.990122 | 1.332785 | |
| std | 9.840844 | 0.003866 | 0.046666 | 0.002946 | 0.002960 | 0.000057 | 0.000590 | 17.192814 | 0.883241 | |
| min | 106.000000 | 0.000000 | 0.000000 | 0.000000 | 0.000000 | 0.000000 | 0.000000 | 12.000000 | 0.200000 | |
| 25% | 126.000000 | 0.000000 | 0.000000 | 0.002000 | 0.000000 | 0.000000 | 0.000000 | 32.000000 | 0.700000 | |
| 50% | 133.000000 | 0.002000 | 0.000000 | 0.004000 | 0.000000 | 0.000000 | 0.000000 | 49.000000 | 1.200000 | |
| 75% | 140.000000 | 0.006000 | 0.003000 | 0.007000 | 0.003000 | 0.000000 | 0.000000 | 61.000000 | 1.700000 | |
| max | 160.000000 | 0.019000 | 0.481000 | 0.015000 | 0.015000 | 0.001000 | 0.005000 | 87.000000 | 7.000000 | |

8 rows × 22 columns
'Distribution of fetal_health:'

| | fetal_health |
|-----|--------------|
| 1.0 | 1655 |
| 2.0 | 295 |
| 3.0 | 176 |

dtype: int64





Before analyzing the results, let's verify if the dataframe has any categorical or null values.

The dataframe has 2126 rows and 22 columns, all of which are of float64 type, indicating that there are no categorical columns. Also, there are no missing values in the dataframe.

The summary statistics show that the 'fetal_health' column has a mean of 1.304 and a standard deviation of 0.614, with values ranging from 1 to 3. The distribution of 'fetal_health' is skewed towards 1,

